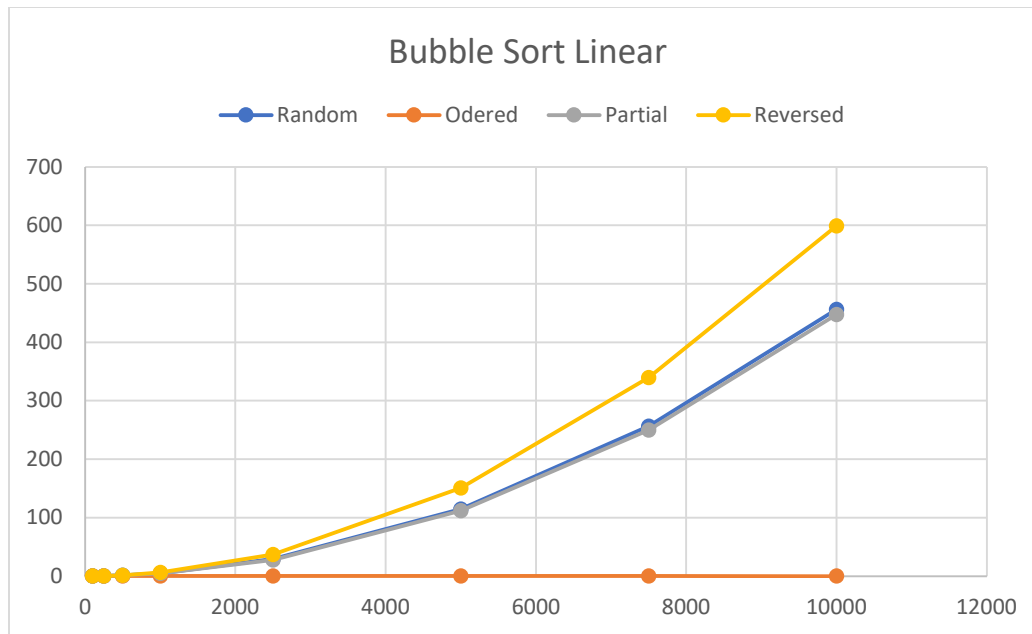
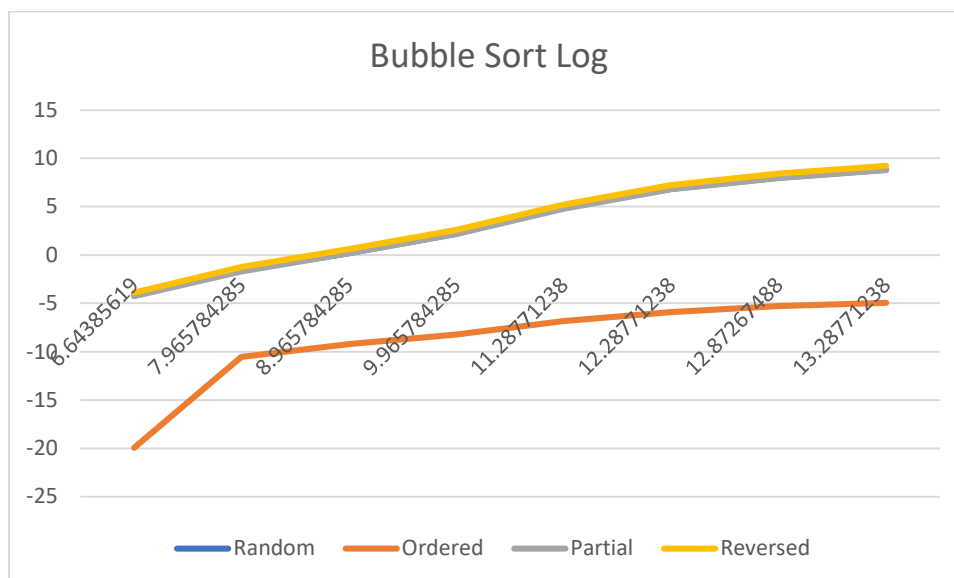




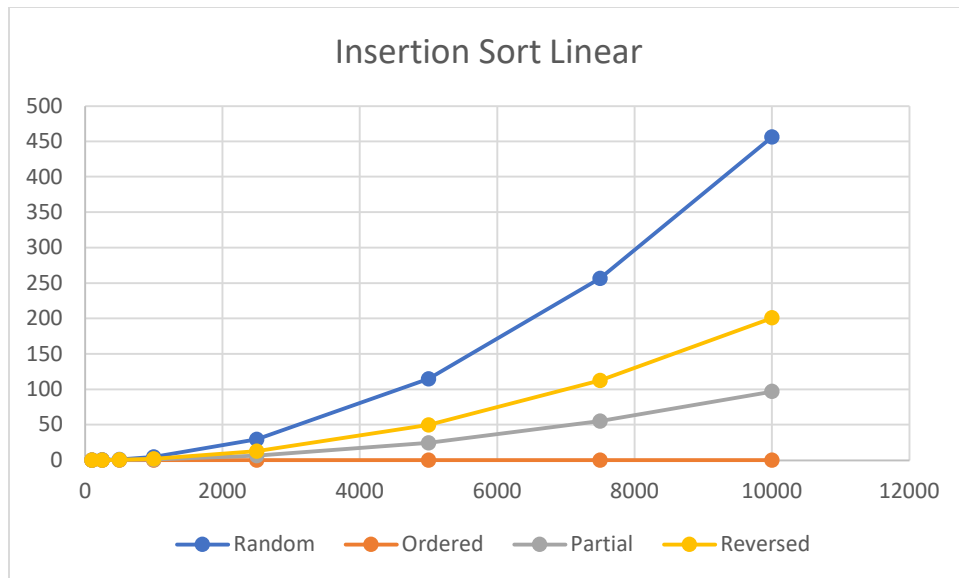
Worst case: for Bubble sort is clearly Reversed which makes sense, because each number must bubble all the way up.

Average case: Partial and Random are both average because depending on the randomness of the list they both have to bubble some far and some very little.

Best case: Ordered is the best case Because all it has to do is run through the list once notice nothing needs moved and it's done.



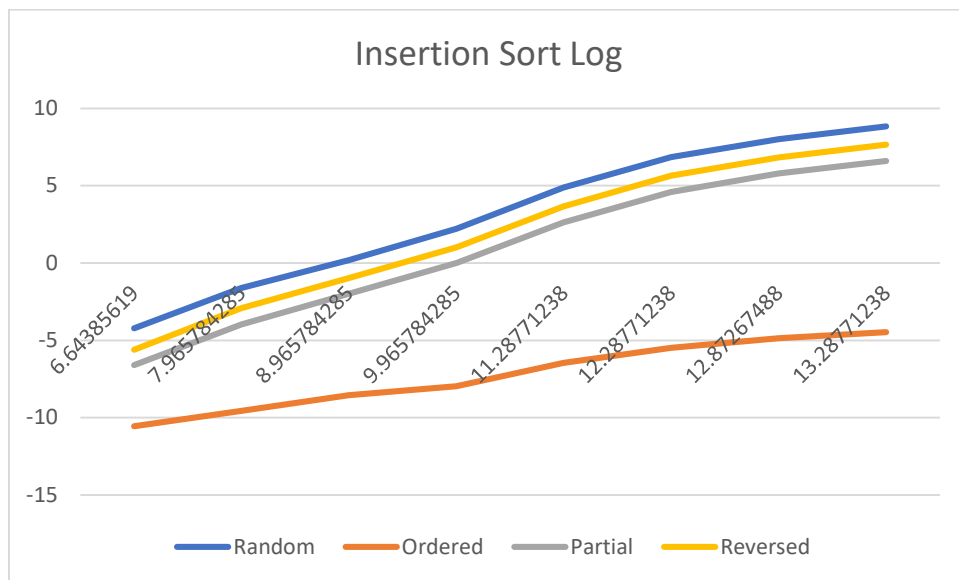
Bubble sort is $O(n^2)$



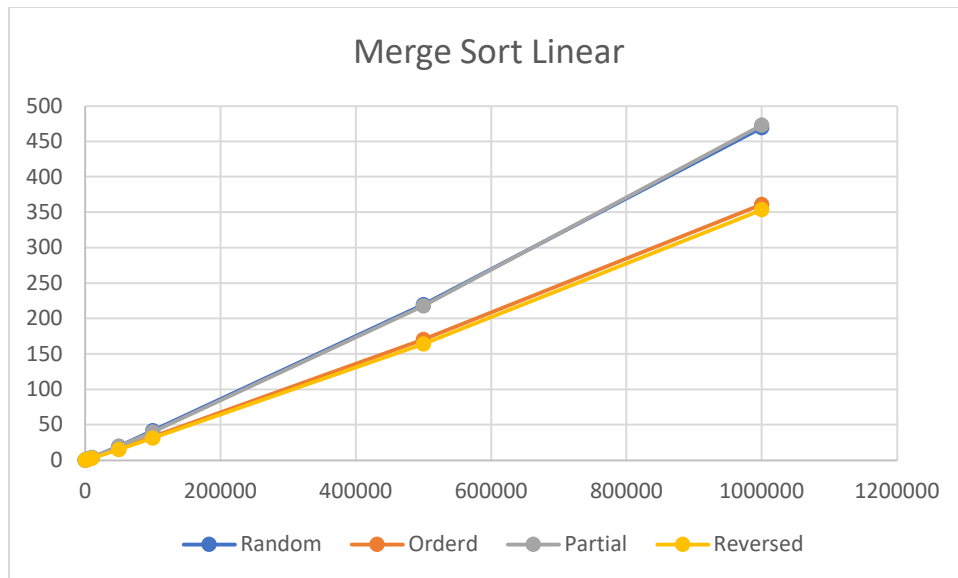
Worst case: Random follows basically the same as bubble sort except that is about twice as good.

Average case: The average is somewhere between partial and reversed. Partial has a bigger difference in insertion because it is able to avoid a lot of unnecessary moves

Best case: Ordered is the best case Because all it has to do is run through the list once notice nothing needs moved and it's done.



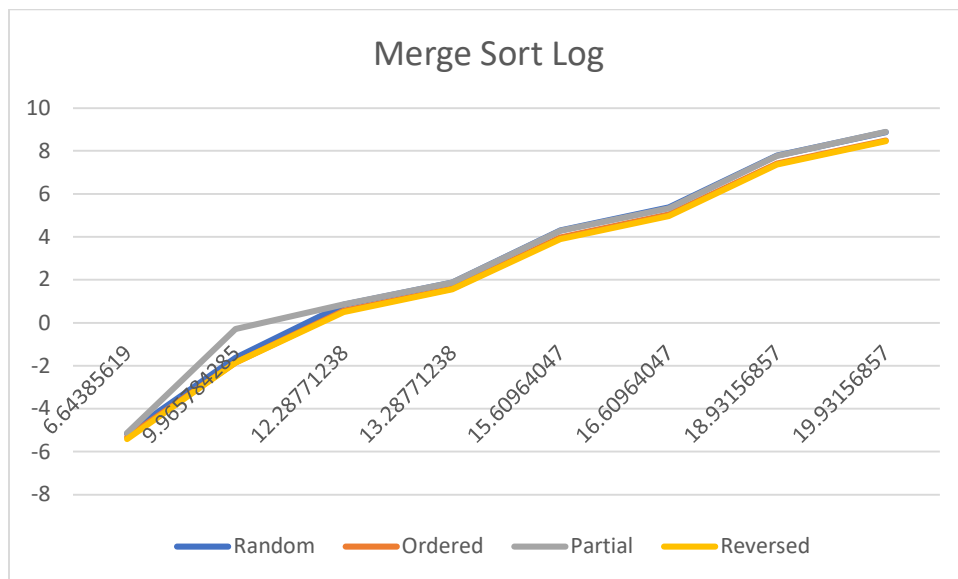
Insertion sort is $O(n^2)$ it is faster than bubble because of the c value.



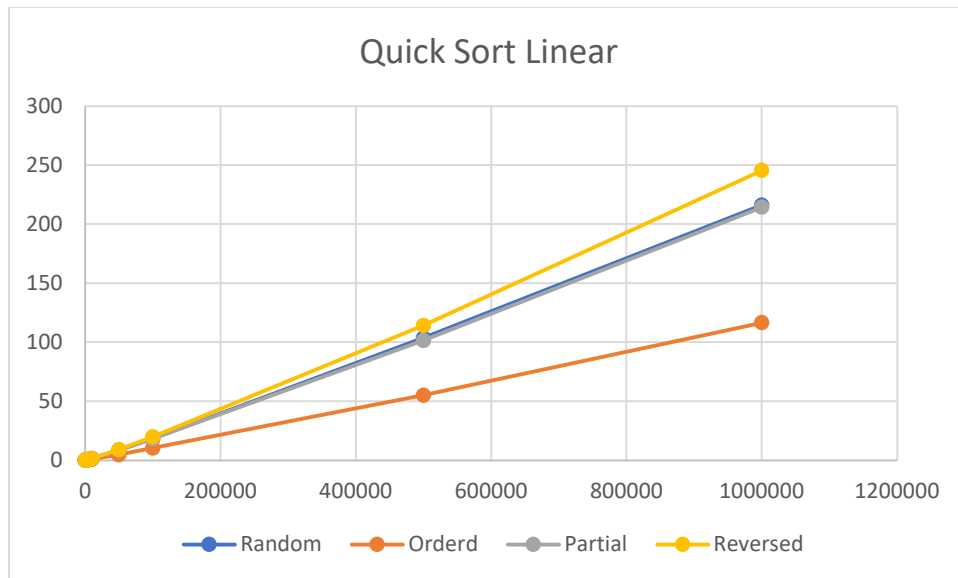
Worst case: In Merge sort random and partial are neck and neck for worst case that is due to the fact that how many recursive steps that have to take place due to the randomness of the arrays.

Average case: There is not really a average case in merge sort.

Best case: Ordered and reversed are neck and neck for best case since all the numbers that are near each other are withing one it takes the same amount of recursive steps to move the and merge the parts together.



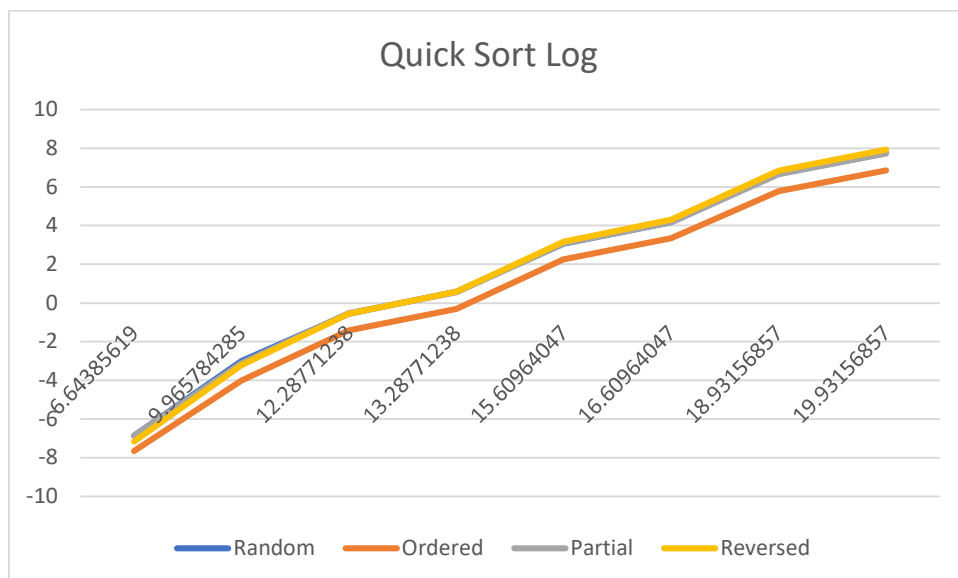
Merge sort is $O(n \log n)$



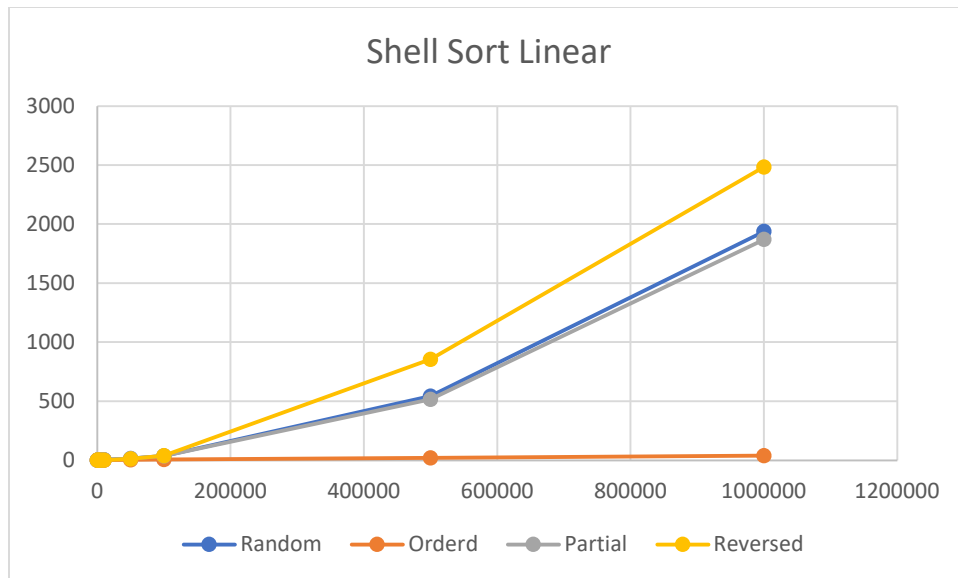
Worst case: The worst case is reversed for quick sort because the pivot is going to have to make the most moves to get everything from one end to the other

Average case: The average case is random and partial due to the randomness of the lists the pivot has to move about the same amount of times

Best case: Ordered is the best for ordered it takes longer than other sorts because the pivot still has to move throughout the whole list at least once rather than other sorting methods being able to do one linear run through the list.



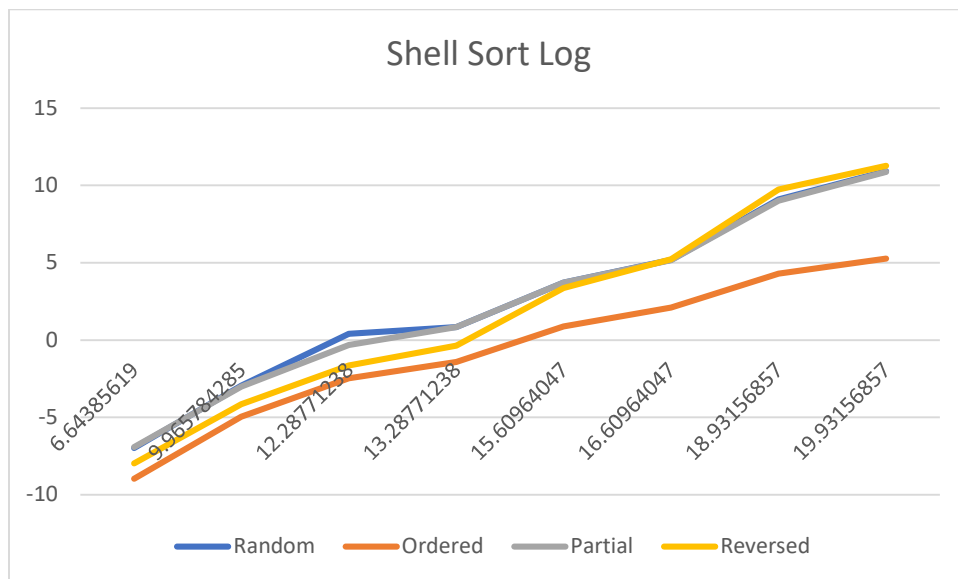
Quick sort is $O(n \log n)$



Worst case: Worst case for Shell sort is reversed because it has to break every part into a new array and combine those arrays.

Average case: Partial and Random are both average because the randomness makes shell sort have to breakdown into less arrays than reversed.

Best case: Ordered is the best case Because all it has to do is run through the list once notice nothing needs moved and it's done.



Shell sort best case is $O(n \log^2(n))$ and worst case is $O(n \log(n^3/2))$.